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MTH603 TODAYQUIZ 2 FILE

SOLVED BY SANAULLAH KHAN

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Final project (MTH620)

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Question # 1 of 10 (Start time: 07:58:05 PM, 19 January 2023) **Total Marks:** 1

6. Let A be an n ×n matrix. The number x is an eigenvalue of A if there exists a non-zero vector v such that Av = xv

Select the correct option

- (●) TRUE
- ( ) FALSE

Click to Save Answer & Move to Next Question

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MTH603 - Numerical Analysis (Quiz # 2) **Quiz Start Time: 07:58 PM, 19 January 2023**

Question # 2 of 10 (Start time: 07:58:14 PM, 19 January 2023) **Total Marks:** 1

$For\ the\ given\ data\ \text{points}\ (1,-3),\ (2,0),\ and\ (3,15),\ the\ first-order\ divide\ difference\ will\ be$

Select the correct option

Reload Math Equations

- ( ) -3
- ( ) -2
- ( ) 2
- (●) 3

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Numerical solution of 2/3 up to four decimal places is _____.

Select the correct option

- 0.6666
- 0.667
- 0.66667
- 0.6667 (selected)

Click to Save Answer & Move to Next Question

7:59 PM 1/19/2023

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Jacobi's method is highly recommended for ..... matrix to compute all the Eigen values and the corresponding eigenvectors.

Select the correct option

- real symmetric (selected)
- non real symmetric
- real un-symmetric
- None of the given choices

Click to Save Answer & Move to Next Question

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Which of the following is a forward difference table for the given values of x and y?

x	0.3	0.7	1.1
y	0.067	0.248	0.518

Select the correct option

Reload Math Equations

x	y	Δy	$\Delta^2 y$
0.3	0.067	0.27	0.089
0.7	0.248	0.181	
1.1	0.518		

x	y	Δy	$\Delta^2 y$
0.3	0.067	0.315	0.1071
0.7	0.248	0.766	
1.1	0.518		

x	y	Δy	$\Delta^2 y$
0.3	0.067	0.089	0.27
0.7	0.248	0.181	
1.1	0.518		

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Reload Math Equations

x	y	Δy	$\Delta^2 y$
0.3	0.067	0.27	0.089
0.7	0.248	0.181	
1.1	0.518		

x	y	Δy	$\Delta^2 y$
0.3	0.067	0.315	0.1071
0.7	0.248	0.766	
1.1	0.518		

x	y	Δy	$\Delta^2 y$
0.3	0.067	0.089	0.27
0.7	0.248	0.181	
1.1	0.518		

x	y	Δy	$\Delta^2 y$
0.3	0.067	0.181	0.089
0.7	0.248	0.27	
1.1	0.518		

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Question # 6 of 10 (Start time: 08:01:35 PM, 19 January 2023) **Total Marks:** 1

Central Difference method is the finite difference method.

Select the correct option

- FALSE
- TRUE

Click to Save Answer & Move to Next Question

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MTH603 - Numerical Analysis (Quiz # 2) **Quiz Start Time: 07:58 PM, 19 January 2023**

Question # 7 of 10 (Start time: 08:01:57 PM, 19 January 2023) **Total Marks:** 1

$For\ the\ given\ data\ \text{points}\ (x_0,y_0),\ (x_1y_1),\ (x_2y_2),\ and\ (x_3,y_3)\ the\ first-order\ divide\ difference\ will\ be\ given\ as$

Select the correct option

Reload Math Equations

- $y[x_0, x_1, x_2]$
- $y[y_0, y_1, y_2]$
- $y[x_0, x_1]$
- $y[x_0]$

Click to Save Answer & Move to Next Question

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While using Jacobi method for the matrix

$$A = \begin{bmatrix} 1 & 2 & \sqrt{2} \\ 2 & 3 & \sqrt{2} \\ \sqrt{2} & \sqrt{2} & 1 \end{bmatrix}$$

the value of 'theta θ' can be found as

Select the correct option

Reload Math Equations

- $\tan 2\theta = \frac{2a_{23}}{a_{22} - a_{33}}$
- (selected) $\tan 2\theta = \frac{2a_{12}}{a_{11} - a_{22}}$
- $\tan 2\theta = \frac{2a_{13}}{a_{11} - a_{33}}$
- $\tan 2\theta = \frac{2a_{13}}{a_{33} - a_{11}}$

Click to Save Answer & Move to Next Question

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Question # 9 of 10 (Start time: 08:02:34 PM, 19 January 2023) **Total Marks:** 1

In Jacobi's Method, the rate of convergence is quite slow compared with other methods.

Select the correct option

- (selected) TRUE
- FALSE

Click to Save Answer & Move to Next Question

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Question # 9 of 10 (Start time: 08:02:34 PM, 19 January 2023) **Total Marks:** 1

In Jacobi's Method, the rate of convergence is quite slow compared with other methods.

Select the correct option

- (●) TRUE
- ( ) FALSE

Click to Save Answer & Move to Next Question

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While using Jacobi method for the matrix

$$A = \begin{bmatrix} 1 & 1/4 & 1/2 \\ 1/4 & 1/3 & 1/4 \\ 1/2 & 1/4 & 1/5 \end{bmatrix}$$

the value of 'theta θ' can be found as

Select the correct option

Reload Math Equations

- ( ) $\tan 2\theta = \frac{2a_{13}}{a_{33} - a_{11}}$
- ( ) $\tan 2\theta = \frac{2a_{23}}{a_{22} - a_{33}}$
- ( ) $\tan 2\theta = \frac{2a_{12}}{a_{11} - a_{22}}$
- (●) $\tan 2\theta = \frac{2a_{13}}{a_{11} - a_{33}}$

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While using Jacobi method for the matrix

$$A = \begin{bmatrix} 1 & 1/4 & 1/2 \\ 1/4 & 1/3 & 1/4 \\ 1/2 & 1/4 & 1/5 \end{bmatrix}$$

the value of 'theta θ' can be found as

Select the correct option

Reload Math Equations

- $\tan 2\theta = \frac{2a_{23}}{a_{22} - a_{33}}$
- $\tan 2\theta = \frac{2a_{13}}{a_{33} - a_{11}}$
- $\tan 2\theta = \frac{2a_{12}}{a_{11} - a_{22}}$
- (selected) $\tan 2\theta = \frac{2a_{13}}{a_{11} - a_{33}}$

Click to Save Answer & Move to Next Question

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Question # 1 of 10 (Start time: 08:08:46 PM, 19 January 2023) **Total Marks:** 1

Exact solution of 2/3 is not exists.

Select the correct option

- (selected) TRUE
- FALSE

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MTH603 - Numerical Analysis (Quiz # 2) **Quiz Start Time: 08:08 PM, 19 January 2023**

Question # 2 of 10 (Start time: 08:09:07 PM, 19 January 2023) **Total Marks:** 1

Given the following data x:0 1 4 y:2 1 4 Value of first order divided difference y[0,1]is

Select the correct option

Reload Math Equations

- 2
- 1
- -2
- -1 (selected)

Click to Save Answer & Move to Next Question

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In Lagrange's interpolation, for the given five points we can represent the function f (x) by a polynomial of degree

Select the correct option

- 6
- 5
- 4 (selected)
- 3

Click to Save Answer & Move to Next Question

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Question # 4 of 10 (Start time: 08:09:36 PM, 19 January 2023) **Total Marks:** 1

By using determinants, we can easily check that the solution of the given system of linear equation exits and it is unique.

Select the correct option

- (●) TRUE
- ( ) FALSE

Click to Save Answer & Move to Next Question

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By using determinants, we can easily check that the solution of the given system of linear equation _____ and it is _____.

Select the correct option

- ( ) nontrivial, inconsistent
- (●) exits, unique
- ( ) exists, consistent
- ( ) trivial, unique

Click to Save Answer & Move to Next Question

8:11 PM
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MTH603 - Numerical Analysis (Quiz # 2) — **Quiz Start Time: 08:08 PM, 19 January 2023**

Question # 7 of 10 (Start time: 08:11:21 PM, 19 January 2023) — **Total Marks:** 1

Given the following data x:1 3 8 y:2 4 9 f(3) can be found by using

Select the correct option

Reload Math Equations

- None
- Newton's backward difference interpolation formula
- Newton's forward difference interpolation formula
- Lagrange's interpolation formula

Click to Save Answer & Move to Next Question

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$|a_{12}|$

be numerically the largest off-diagonal element of A, then we can construct orthogonal matrix S1 by Jacobi's method as

Select the correct option

Reload Math Equations

- $\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- $\begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\cos\theta \\ 0 & \sin\theta & -\sin\theta \end{bmatrix}$

Click to Save Answer & Move to Next Question

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$|a_{13}|$

be numerically the largest off-diagonal element of A, then we can construct orthogonal matrix S1 by Jacobi's method as

Select the correct option

Reload Math Equations

- $\begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\cos\theta \\ 0 & \sin\theta & -\sin\theta \end{bmatrix}$
- $\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

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Question # 2 of 10 (Start time: 08:14:48 PM, 19 January 2023) **Total Marks:** 1

For the giev two data points , *the* degree *of Lagrange's* interpolation *polynomial could be*

x	0.3	0.7
y	0.067	0.248

Select the correct option

Reload Math Equations

- four
- two
- one
- three

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Question # 4 of 10 (Start time: 08:15:16 PM, 19 January 2023) **Total Marks:** 1

Which of the following is a forward difference table for the given values of x and y?

x	0.1	0.5	0.9
y	0.003	0.148	0.37

Select the correct option

Reload Math Equations

x	y	Δy	$\Delta^2 y$
0.1	0.003	0.287	0.644
0.5	0.148	0.357	
0.9	0.37		

x	y	Δy	$\Delta^2 y$
0.1	0.003	0.145	0.077
0.5	0.148	0.222	
0.9	0.37		

x	y	Δy	$\Delta^2 y$
0.1	0.003	0.567	0.704
0.5	0.148	0.137	
0.9	0.37		

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Question # 5 of 10 (Start time: 08:15:50 PM, 19 January 2023) **Total Marks:** 1

While using Jacobi method for the matrix

$$A = \begin{bmatrix} 1 & \sqrt{3} & \sqrt{2} \\ \sqrt{3} & 3 & 2 \\ \sqrt{2} & 2 & 1 \end{bmatrix}$$

and theta θ=0.5535 the orthogonal matrix S1 will be given as

Select the correct option

Reload Math Equations

$$S_1 = \begin{bmatrix} \cos 0.5535 & -\sin 0.5535 & 0 \\ \sin 0.5535 & \cos 0.5535 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 0.5535 & -\sin 0.5535 \\ 0 & \sin 0.5535 & \cos 0.5535 \end{bmatrix}$$

$$S_1 = \begin{bmatrix} \cos 0.5535 & 0 & 0 \\ 0 & -\sin 0.5535 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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$$A = \begin{bmatrix} \sqrt{3} & 3 & 2 \\ \sqrt{2} & 2 & 1 \end{bmatrix}$$

and theta θ=0.5535 the orthogonal matrix S1 will be given as

Select the correct option

Reload Math Equations

- $S_1 = \begin{bmatrix} \cos 0.5535 & -\sin 0.5535 & 0 \\ \sin 0.5535 & \cos 0.5535 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- (selected) $S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 0.5535 & -\sin 0.5535 \\ 0 & \sin 0.5535 & \cos 0.5535 \end{bmatrix}$
- $S_1 = \begin{bmatrix} \cos 0.5535 & 0 & 0 \\ 0 & -\sin 0.5535 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- $S_1 = \begin{bmatrix} \cos 0.5535 & 0 & -\sin 0.5535 \\ 0 & 1 & 0 \\ \sin 0.5535 & 0 & \cos 0.5535 \end{bmatrix}$

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Question # 9 of 10 (Start time: 08:16:08 PM, 19 January 2023) Total Marks: 1

While using Jacobi method for the matrix

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

the value of 'theta θ' can be found as

Select the correct option

Reload Math Equations

- $\tan 2\theta = \frac{2a_{13}}{a_{33} - a_{11}}$
- $\tan 2\theta = \frac{2a_{13}}{a_{11} - a_{33}}$
- $\tan 2\theta = \frac{2a_{12}}{a_{11} - a_{22}}$
- (selected) $\tan 2\theta = \frac{2a_{23}}{a_{22} - a_{33}}$

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$|a_{23}|$

be numerically the largest off-diagonal element of A, then we can construct orthogonal matrix S1 by Jacobi's method as

Select the correct option

Reload Math Equations

- $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\cos\theta \\ 0 & \sin\theta & -\sin\theta \end{bmatrix}$
- $\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- $\begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$
- (●) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$

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If any ten data points are given,the degree of Lagrange's interpolation polynomial could be

Select the correct option

- (●) nine
- twelve
- eleven
- ten

Click to Save Answer & Move to Next Question

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Question # 2 of 10 (Start time: 08:25:20 PM, 19 January 2023) **Total Marks:** 1

For the given four data points, the degree of Lagrange's interpolation polynomial could be

x	0.3	0.7	0.9	1.0
y	0.067	0.248	0.518	0.6812

Select the correct option

Reload Math Equations

- five
- four
- six
- three (selected)

Click to Save Answer & Move to Next Question

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While using Jacobi method for the matrix

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

the value of 'theta θ' can be found as

Select the correct option

Reload Math Equations

- $\tan 2\theta = \frac{2a_{13}}{a_{33} - a_{11}}$
- $\tan 2\theta = \frac{2a_{23}}{a_{22} - a_{33}}$
- $\tan 2\theta = \frac{2a_{13}}{a_{11} - a_{33}}$ (selected)
- $\tan 2\theta = \frac{2a_{12}}{a_{11} - a_{22}}$

Click to Save Answer & Move to Next Question

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MTH603 - Numerical Analysis (Quiz # 2) — **Quiz Start Time: 08:23 PM, 19 January 2023**

Question # 5 of 10 (Start time: 08:27:11 PM, 19 January 2023) — **Total Marks:** 1

$For\ the\ given\ data\ \text{points}\ (4, 45),\ (5, 104),\ and\ (6, 190),\ the\ \text{first} - order\ divide\ difference\ will\ be$

Select the correct option

Reload Math Equations

- none
- 76
- 82
- 59 (selected)

Click to Save Answer & Move to Next Question

8:27 PM 1/19/2023

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MTH603 - Numerical Analysis (Quiz # 2) — **Quiz Start Time: 08:23 PM, 19 January 2023**

Question # 6 of 10 (Start time: 08:27:29 PM, 19 January 2023) — **Total Marks:** 1

What will be the value of first order divided difference f[1,5]for the following data x:0 1 5 y:2 1 5

Select the correct option

Reload Math Equations

- 3
- 0
- 1 (selected)
- 2

Click to Save Answer & Move to Next Question

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Question # 7 of 10 (Start time: 08:27:49 PM, 19 January 2023) **Total Marks:** 1

While using Jacobi method for the matrix

$$A = \begin{bmatrix} 1 & \sqrt{3} & 2 \\ \sqrt{3} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{bmatrix}$$

and 'theta θ =pi/4',the orthogonal matrix S1 will be given by

Select the correct option

Reload Math Equations

- $S_1 = \begin{bmatrix} \cos\frac{\pi}{4} & 0 & -\sin\frac{\pi}{4} \\ 0 & 1 & 0 \\ \sin\frac{\pi}{4} & 0 & \cos\frac{\pi}{4} \end{bmatrix}$
- $S_1 = \begin{bmatrix} \cos\frac{\pi}{4} & 0 & 0 \\ 0 & -\sin\frac{\pi}{4} & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- $S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\frac{\pi}{4} & -\sin\frac{\pi}{4} \\ 0 & \sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{bmatrix}$

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Power method is applicable if the eigen values are _____.

Select the correct option

- negative and distinct
- real and distinct
- real and equal
- positive and distinct

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Question # 9 of 10 (Start time: 08:28:18 PM, 19 January 2023) **Total Marks:** 1

For the giev three data points, *the* degree *of Lagrange's* interpolation *polynomial could be*

x	0.3	0.7	0.9
y	0.067	0.248	0.518

Select the correct option

Reload Math Equations

- Three
- Two
- Five
- Four

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Question # 10 of 10 (Start time: 08:28:28 PM, 19 January 2023) **Total Marks:** 1

An eigenvector V is said to be normalized if the coordinate of largest magnitude is equal to _____.

Select the correct option

- zero
- unity

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Question # 1 of 10 (Start time: 08:34:34 PM, 19 January 2023) **Total Marks:** 1

Power method is applicable if the eigen values are real and distinct.

Select the correct option

- (●) TRUE
- ( ) FALSE

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MTH603 - Numerical Analysis (Quiz # 2) **Quiz Start Time: 08:34 PM, 19 January 2023**

Question # 2 of 10 (Start time: 08:34:48 PM, 19 January 2023) **Total Marks:** 1

Given the following data x:1 2 7 11 y:6 10 13 37 Which formula is useful in finding the interpolating polynomial?

Select the correct option

Reload Math Equations

- ( ) Newton's backward difference interpolation formula
- (●) Lagrange's interpolation formula
- ( ) None
- ( ) Newton's forward difference interpolation formula

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Question # 3 of 10 (Start time: 08:35:02 PM, 19 January 2023) **Total Marks:** 1

The dominant eigenvector of a matrix is an eigenvector corresponding to the eigenvalue of largest magnitude (for real numbers, smallest absolute value) of that matrix.

Select the correct option

- TRUE
- FALSE

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MTH603 - Numerical Analysis (Quiz # 2) **Quiz Start Time: 08:34 PM, 19 January 2023**

Question # 4 of 10 (Start time: 08:35:11 PM, 19 January 2023) **Total Marks:** 1

$For\ the\ given\ data\ \text{points}\ (1,-3),\ (2,0),\ and\ (3,15),\ the\ zero-order\ divide\ difference\ will\ be$

Select the correct option

Reload Math Equations

- -2
- -3
- 0
- -1

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MTH603 - Numerical Analysis (Quiz # 2) — **Quiz Start Time: 08:34 PM, 19 January 2023**

Question # 5 of 10 (Start time: 08:35:30 PM, 19 January 2023) — **Total Marks:** 1

Given the following data x:0 1 4 y:2 1 4 Value of first order divided difference y[0,1]is

Select the correct option

Reload Math Equations

- -1
- 1
- 2
- -2

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MTH603 - Numerical Analysis (Quiz # 2) — **Quiz Start Time: 08:34 PM, 19 January 2023**

Question # 6 of 10 (Start time: 08:35:46 PM, 19 January 2023) — **Total Marks:** 1

Let $[A]$ be a 3×3 real symmetric matrix with $|a_{23}|$ be the numerically largest off − diagonal element then using Jacobi's method the value of θ can be found by

Select the correct option

Reload Math Equations

- $\tan 2\theta = \frac{2a_{23}}{a_{22} - a_{33}}$
- $\tan 2\theta = \frac{2a_{13}}{a_{33} - a_{11}}$
- $\tan 2\theta = \frac{2a_{13}}{a_{11} - a_{33}}$
- $\tan 2\theta = \frac{2a_{12}}{a_{22} - a_{11}}$

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MTH603 - Numerical Analysis (Quiz # 2) Quiz Start Time: 08:34 PM, 19 January 2023

Question # 6 of 10 (Start time: 08:35:46 PM, 19 January 2023) **Total Marks:** 1

Let $[A]$ be a 3×3 real symmetric matrix with $|a_{23}|$ be the numerically largest off − diagonal element then using Jacobi's method the value of θ can be found by

Select the correct option

Reload Math Equations

- ◉ $\tan 2\theta = \frac{2a_{23}}{a_{22} - a_{33}}$
- ○ $\tan 2\theta = \frac{2a_{13}}{a_{33} - a_{11}}$
- ○ $\tan 2\theta = \frac{2a_{13}}{a_{11} - a_{33}}$
- ○ $\tan 2\theta = \frac{2a_{12}}{a_{22} - a_{11}}$

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Question # 7 of 10 (Start time: 08:36:39 PM, 19 January 2023) **Total Marks:** 1

If any three data points are given, the formula for Lagrange's interpolation polynomial will be

Select the correct option

Reload Math Equations

- ○ $y = f(x) = \frac{(x-x_1)(x-x_2)}{(x_1-x_0)(x_1-x_2)}y_0 + \frac{(x-x_0)(x-x_2)}{(x_0-x_1)(x_0-x_2)}y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}y_2$
- ○ $y = f(x) = \frac{(x_0-x_1)(x_0-x_2)}{(x-x_1)(x-x_2)}y_0 + \frac{(x_1-x_0)(x_1-x_2)}{(x-x_0)(x-x_2)}y_1 + \frac{(x_2-x_0)(x_2-x_1)}{(x-x_0)(x-x_1)}y_2$
- ◉ $y = f(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}y_2$
- ○ $y = f(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}y_2 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}y_0$

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MTH603 - Numerical Analysis (Quiz # 2) — **Quiz Start Time: 08:34 PM, 19 January 2023**

Question # 8 of 10 (Start time: 08:37:06 PM, 19 January 2023) — **Total Marks:** 1

For the given data points $(x_0 y_0)$, $(x_1 y_1)$, $(x_2 y_2)$, *and* $(x_3 y_3)$ *the zero − order divide difference will be given as*

Select the correct option

Reload Math Equations

- (●) $y[x_0]$
- ( ) $y[x_0, x_1]$
- ( ) $y[y_0, y_1]$
- ( ) $y[y_0]$

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Question # 9 of 10 (Start time: 08:37:16 PM, 19 January 2023) — **Total Marks:** 1

The Power method can be used only to find the eigenvalue of A that is largest in absolute value—we call this eigenvalue the dominant eigenvalue of A.

Select the correct option

- (●) TRUE
- ( ) FALSE

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Question # 9 of 10 (Start time: 08:37:16 PM, 19 January 2023) **Total Marks:** 1

The Power method can be used only to find the eigenvalue of A that is largest in absolute value—we call this eigenvalue the dominant eigenvalue of A.

Select the correct option

- (●) TRUE
- ( ) FALSE

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Question # 10 of 10 (Start time: 08:37:58 PM, 19 January 2023) **Total Marks:** 1

The Jacobi iteration _____, if A is strictly diagonally dominant.

Select the correct option

- ( ) diverges
- (●) converges

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